

Chapter 5

Worksheet

Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

Worksheet for Chapter 5: Introduction to Basic Analytics

Instructions: Complete each section of this worksheet individually or in small groups as directed by your teacher. Use a calculator, notebook, or digital tools (like Google Sheets) if needed for calculations or visuals. Be prepared to discuss your answers in class or share your mini project findings. Have fun becoming a data analyst!

Part 1: Understanding Analytics (15 Points)

Objective: Grasp the concept of analytics and its difference from data through real-life scenarios and reflection.

- **Define and Differentiate (10 Points)** a. **What is Analytics? (5 Points):** In 2-3 sentences, explain what analytics means in your own words. (Hint: Think of it as solving a mystery with data.)
- b. **Data vs. Analytics (5 Points):** Describe the difference between data and analytics using an example from your life (e.g., tracking screen time). Write 2-3 sentences.
- **Real-Life Connection (5 Points)** Think of one app or website you use daily (like Instagram or Netflix). How do you think analytics works behind the scenes to personalize your experience? Write 2-3 sentences.

Part 2: Asking the Right Questions (20 Points)

Objective: Practice framing specific, data-driven questions and explore the role of curiosity in analytics.

- **Question Makeover (10 Points)** Below are two vague questions. Rewrite each into a specific, measurable question that could guide data analysis. Write one sentence explaining why your new question is better. a. **Vague Question:** Why do I feel tired all the time?
 - **New Question (3 Points):** _____
 - **Why Better? (2 Points):** _____ b. **Vague Question:** Why isn't my school club popular?
 - **New Question (3 Points):** _____
 - **Why Better? (2 Points):** _____
- **Curiosity Spark (10 Points)** Think of something in your school or personal life you're curious about (e.g., "Why do I do better in some subjects?"). a. **Your Wonder (5 Points):** Write down your big curious question. Then, break it into 2 smaller, specific questions you could analyze with data.
 - Big Question: _____
 - Smaller Question 1: _____
 - Smaller Question 2: _____ b. **Why Curious? (5 Points):** In 2-3 sentences, explain why this question matters to you and how answering it with data could help.

Part 3: Hands-On Mini Analytics Project (25 Points)

Objective: Apply analytics by analyzing a small dataset, summarizing results, and suggesting actions.

- **Analyze Class Data (20 Points)** Use this sample dataset of math test scores for 10 students: 65, 70, 85, 90, 55, 80, 75, 60, 88, 72. (If you have real class data from your teacher, use that instead!) Your question: "What's the typical score in this class, and are there students who might need help?" a. **Calculate Stats (10 Points):** Show your work for each.
 - Mean (Average Score, 4 Points): _____ Answer: _____
 - Median (Middle Score after sorting, 3 Points): Sorted List: _____ Answer: _____
 - Range (Highest minus Lowest, 3 Points): _____ Answer: _____b. **Spot Patterns (5 Points):** Look at the data. Is there an outlier (unusual score)? What pattern do you notice about most scores? Write 2-3 sentences.
- c. **Suggest Action (5 Points):** Based on your findings, suggest one action the teacher or students could take (e.g., extra help for low scores). Write 2 sentences.
- **Quick Visual (5 Points)** On a separate sheet or below, sketch a simple bar chart for the test scores (x-axis: Student 1-10, y-axis: Score 0-100). Label the axes. Write one sentence about what the chart shows.

Part 4: Creative Application (10 Points)

Objective: Connect analytics to personal interests through a mini experiment plan.

- **Personal Analytics Plan (10 Points)** Choose something you can track over 5 days (e.g., hours of sleep, steps walked, or mood rating 1-10). Answer the following:
 - What will you track? _____
 - What question are you asking? (e.g., "Does sleep affect my mood?") _____
 - How will you record data (notebook, app)? _____
 - Predict one pattern or finding you might see. _____
 - After tracking (or imagining data if time is short), write one conclusion and one action in 2-3 sentences.

Part 5: Reflection (5 Points)

Objective: Think critically about the value of analytics in everyday life.

- **Think Big (5 Points)** How can analytics help you in school, hobbies, or personal goals? Write 2-3 sentences with a specific example (e.g., improving grades or gaming skills).